

Artifact Status

Paper: “Suborbital and Orbital Real-Time Localization of Gamma-Ray Bursts via Utility-Guided Coordination”

Intended badges

The authors intend to apply for the following EMSOFT 2026 artifact badges:

- **Available**
- **Reviewed**
- **Reproducible**

Rationale

Available

The artifact consists of two persistently archived components:

1. the source code, environment specifications, precomputed results, analysis notebook, documentation, and accepted paper version, archived as a version-specific snapshot of the artifact repository; and
2. the large detector models and transient datasets, distributed separately through Zenodo with a published checksum.

The corresponding persistent identifiers are:

Artifact DOI: <https://doi.org/10.5281/zenodo.21572120>

Artifact Dataset DOI: <https://doi.org/10.5281/zenodo.21497891>

Continued development will take place in the following GitHub repository:

GitHub repository: <https://github.com/Daisy0419/GRB-Search-Pipeline>

Reviewed

The artifact provides:

- a short path that regenerates Figures 4–11 from the distributed results;
- a complete path that regenerates the training and evaluation results; and
- an optional path for rerunning the platform-sensitive mapping benchmark.

The required commands, inputs, output locations, and expected outputs are documented in `README.md`.

Reproducible

The artifact includes the author-created code, methods, environment specifications, simulated inputs, and analysis workflow needed for an independent evaluator to regenerate the computational results. The complete workflow is documented in README Sections 1–4.

Scope and limitations

- Figures 1–3 are explanatory diagrams rather than experiment outputs.
- Exact timing values are platform-sensitive. The reference hardware is specified in `REQUIREMENTS` and `README.md`.
- Sections 1–3 require Linux ARM64 (`aarch64`).
- The optional Intel portion of Figure 4 requires Linux x86-64.
- The full experiment is computationally expensive; approximate runtimes are listed in `REQUIREMENTS`.